

The category axioms, and what defines them in the Frobenius calculus

Freyd, Categories, Algebras §2.11-§2.12 against functorialSemanticsForRelationalTheories.pdf Def. 4.1

Lean: Freyd/S2_10.lean (Allegory), diag/CB.lean (CartBicat), diag/CB.Allegory.lean (allegoryofCartBicat)

Where the pictures come from. Every picture of a theorem is emitted by `!scripts/diag-export` from that theorem's Lean script. Nothing about a statement is redrawn here. The two figures marked *transcribed* are the exception: they copy the paper's definition pictures, and a definition has no statement to export.

Equation numbering. The numbers below are as of 2017 preprint's. The CartBicat fields no longer encode them – a field is named for what it says, « $\Delta \leq 1$ » rather than $\Delta \leq 1$, 37 – and each dovetail carries the preprint number. The repo's copy of the paper is now the published version – Poly. J. Math. 2 (6), 20 Oct 2025, doi:10.6083/polyjmath.2.6 – which remembers. Translation, published – preprint (34) – (37), (35) – (38), (36) – (39), (37) – (40), (38) – (41), (39) – (42), (40) – (43), (39) – (35), and Example 2.30 (3) (5), (14) – (11), (12), the preprint (38) is unnumbered there. Pages: Def. 4.1 at 20–21 (two 17), the Γ notation at 21 (two 19), Lemma 4.2 at 21–22 (two 19). Remembering the Lean fields is a separate decision and has not been made.

Freyd's allegory takes reciprocation $\circ$ and intersection $\cap$ as **primitive operations** and lists eight **equational axioms** on them. A cartesian bicategory of relations takes neither: every object carries one special Frobenius structure $\Delta, \nabla, \tau, \iota$, the ham-mond $\leq$ is primitive, and both $\circ$ and $\cap$ are **defined** from the four dots. Every one of Freyd's eight axioms is then a theorem. The table below is that dictionary, one row per axiom.

Reading the pictures: left to right is composition – the book's diagram order. R, S means first R then S – vertical stacking is $\circ$, a bare wire is an identity, a white box is a relation, and a black box that is mirrored and tinted is its converse – the mirroring is the paper's own notation, the tint is this note's, because at a glance a chamber is easy to miss. The label inside is **not mirrored**: the paper can reverse a single letter and still be read, a whole term like $R \circ S$ cannot. All four dots are solid, as Def. 4.1 draws them. The exporter draws a meet or a converse as a **labelled** box, so a composite under $\circ$ appears as a single box reading $R \circ S$; $S \circ \Delta$ draws two stack runs; and any subterm with no shape in this calculus – an associator, say – deserves to an opaque box. The column words write the converse R^* , Freyd's symbol, where the paper writes R° ; these terms are read against the book's theorems and one symbol per idea beats asking the reader to match two. The Lean sources write R^* as well. The paper reserves $\circ$ for the colour swap of its §7, but §7 is a worked example in other models – see the caution at the end of this note – and nothing here formalises it, so in a repo that is $\text{Rel}(\text{Set})$ throughout there is no clash to protect against.

In a chain table the symbol at the head of each row is the relation that **step was proved**: $\Leftarrow$ for a reshaping, $\Leftarrow$ for a genuine inequality. Counting them is the point of drawing a proof at all – a chain of $\Leftarrow$ s between equalities says the theorem is one law and a lot of bookkeeping.

1. Definition 4.1, as the paper states it

functorialSemanticsForRelationalTheories.pdf Def. 4.1, pp. 20–21 – transcribed here as `Class CartBicat (diag/CB.lean)`, field for field. Every picture below is exported from the Lean field, so the class and the definition cannot drift apart.

A cartesian bicategory of relations is a poset-enriched category, symmetric monoidal, satisfying four conditions. Clauses 1 and 2 are the two structures every object carries; clause 3 makes each the other's adjoint and imposes Frobenius; clause 4 is the condition on arrows.

The tower this note describes is STRICT: the two bracketings of a threefold product are the same object, so no associator or unitor appears in any picture. The coherence arrows the paper suppresses in its own diagrams are absent here because they do not exist.

1.1. Every object carries a cocommutative comonoid (Δ, ∇)

	(8) cocommutativity <code>CartBicat.Δ_assoc</code> – copying twice on the left is copying twice on the right. (9) cocommutativity <code>CartBicat.Δ_comm</code> – the two copies are interchangeable. (10) counit <code>CartBicat.Δ_counit</code> – copy, then discard one copy, and nothing has happened.
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1.2. Every object carries a commutative monoid (τ, ι)

	(5) associativity <code>CartBicat.τ_assoc</code> (6) commutativity <code>CartBicat.τ_comm</code> (7) unit <code>CartBicat.τ_unit</code> – create, then merge, and nothing has happened.
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1.3. Five inequations relate the monoid and the comonoid

	(37) $\nabla : \Delta \leq 1$ – half of $\Delta \leq \nabla$
	(38) $1 \leq \Delta : \nabla$ – the other half
	(39) $! : 1 \leq !$ – half of $! \leq ?$
	(40) $! \leq ! : ?$ – the other half
	(41) the Frobenius law <code>CartBicat.frob_left</code> and its mirror <code>frob_right</code> . This is the one equation of the group; the other four are inequalities.

The special law $\Delta : \nabla = 1$ is NOT part of the definition. The paper derives it – “one direction is given by (38) and the other is proved as follows” – and so does the Lean, as `Freyd.Diag.CartBicat.special`; its picture is in the correspondence table below.

1.4. Every arrow is a lax comonoid homomorphism

	(42) $R : \Delta \leq \Delta : (R \otimes R)$ <code>CartBicat.lax_Δ</code> – running R once and copying is below copying and running it on both strands.
	(43) $R : ! \leq !$ <code>CartBicat.lax_!</code>

Read as inequalities in one direction only: an arrow that satisfied these as EQUATIONS would be a map, and the whole point is that a general relation is not.

2. Freyd's primitives are the Frobenius calculus's definitions

Freyd's primitive, and what defines it here	picture
reciprocal $R^* = \$2.11$, primitive. Here <code>conv R := bend (R * 1) ; cap (diag/CB.lean:223)</code> , not a generator. Input arrives bottom left and runs under R ; the cap ∇ at the top left creates a pair, one strand feeding R and the other leaving as the output at the top right; the cap ∇ at the bottom right annihilates R 's output against the input. It is the cap and the cap that reverse the arrow – a wire merely bumped up over the box and back down is an isotopy of R itself.	
intersection $R \cap S = \$2.11$, primitive. Here the convolution <code>meet R S := Δ ; (R * S) ; ∇ (diag/CB.Derived.lean:24)</code> . Copy the input, run both, merge; the merge forces the two results to agree.	
containment $R \sqsubseteq S = \$2.11$, defined as $R \cap S = R$. Here the poset-enrichment $\sqsubseteq$ is swapped and $\cap$ is derived, so the two directions are intuitive. <code>meet_eq_left_of_le (diag/CB.Derived.lean:196)</code> is the bridge.	no picture – this is the $\circ$ -cell itself
maximal morphism $\tau = \$2.15$, and <code>Freyd</code> needs a unitary allegory for it. Here <code>top a b := ! ; ? (diag/CB.Derived.lean:28)</code> is free in Def. 4.1. <code>Freyd.Diag.«s_top»</code>	

3. The eight axioms, and what proves each

Freyd's axiom, and what supplies it	picture, from the Lean statement
(R^*) * $= R$ – $\$2.11$ <code>recip_recip</code> , here the snake, from (41). <code>Freyd.Diag.CartBicat.conv_conv</code> . Bending down and back up straightens out, so bending twice is the identity.	
($R S$) * $= S^* R^*$ – $\$2.11$ <code>recip_comp</code> , here the same mirroring. <code>Freyd.Diag.CartBicat.conv_conv</code> . Reflecting a chain left to right reverses it; each box comes back flipped.	
($R \cap R$) * $= R$ – $\$2.11$ <code>inter_idem</code> , here (42) and the special law. <code>Freyd.Diag.meet_idem</code> . Copy, run R twice, merge – (42) pushes R back through the copy and Δ ; ∇ at 1 closes the bubble.	
($R \cap S$) * $= S \cap R$ – $\$2.11$ <code>inter_comm</code> , here cocommutativity (9) and commutativity (6). <code>Freyd.Diag.meet_comm</code> . The two strands of Δ are interchangeable, and so are those of ∇ .	
($R \cap (S \cap T)$) * $= (R \cap S) \cap T$ – $\$2.11$ <code>inter_assoc</code> , here cocommutativity (8) and associativity (5). <code>Freyd.Diag.meet_assoc</code> . Either bracketing builds the same three-way copy tree, and dually the same merge tree.	
semi-distributivity $R (S \cap T) \sqsubseteq (R S) \cap T$ – $\$2.11$ <code>semidistrib</code> , $\$2.136$, here (42), the lax comonoid inequation. <code>Freyd.Diag.semidistrib_of_lax</code> . Running R on each strand, copying is below copying and then running R on each strand.	
modular identity $R S \cap T \sqsubseteq (R \cap S) T$ – $\$2.11$ <code>modular</code> , the axiom Freyd adjoints “we adjoint one more containment, the law of modularity”. Here (41) and (42), derived. <code>Freyd.Diag.modular_of_frobenius</code> . Strip the prefex Δ ; ($R \otimes T$) of both sides and what is left is $\nabla (S \text{ slide_conv})$ (<code>diag/CB.lean:560</code>) – a box slides through a merge at the cost of its converse on the other strand.	

4. The other direction: what each law of Def. 4.1 says in Freyd's language

picture, from the Lean statement	Def. 4.1	Freyd
	(37) $\nabla : \Delta \leq 1$ half of $\Delta \leq \nabla$	$\Delta \leq 1$ – Δ is simple $\$2.13$. Not used anywhere in the bridge; see below.
	(38) $1 \leq \Delta : \nabla$ the other half of $\Delta \leq \nabla$	$1 \leq \Delta$ – Δ is a centre $\$2.13$ with (37), Δ is a map.
	(39) $! : 1 \leq !$ half of $! \leq ?$	$! : 1 \leq !$ – $!$ is simple $\$2.13$. Also unused in the bridge.
	(40) $! \leq ! : ?$ the other half of $! \leq ?$	$! \leq !$ – $!$ is a centre $\$2.13$ with (39), $!$ is a map. This what makes $! \Rightarrow !$; $?$ contain every arrow.
	(41) the Frobenius law an axiom. $! \otimes \Delta$ and $\nabla \otimes R$ draw as stacked runs; only the associator $\circ$ stays an opaque box, having no shape in this calculus.	the modular identity $\$2.112$ – and reciprocation $\$2.11$ itself, since the cap and copy only satisfy the snake because of (41).
	(42) every arrow is a lax comonoid homomorphism	$\Delta \Delta^* \dashv \vdash \Delta$ – Δ is injective , so by $\$2.141$ at $\tau = \Delta$ is Δ is a monic map. In $\text{In}(\text{Set})$: copying is total and loses nothing.
	(43) every arrow is lax for the counit	$\$2.136$'s $R (S \cap T) \sqsubseteq (R S) \cap T$, read as $S \sqsubseteq \cap S^*, T \sqsubseteq \cap S^*$ – Freyd's semi-distributivity, product-free. $\$2.152$'s step “ p_a is maximal in (α, λ) ”, hence $R \circ p_a \leq p_a$ – that is, $R \circ \tau$.

5. Unitary and pre-tabular – the two conditions the converse needs

Freyd's §2.11 axioms are all a cartesian bicategory has to give back. Coing the other way needs two further conditions, and they are exactly the two repo has not formalised. Both definitions below are quoted from `Freyd/S2_10.lean`, which follows the book.

condition	Freyd	what the Frobenius side already has
unitary $\$2.115$	T is a partial unit when $\text{PartUnit } T$ is the largest endomorphism on it, $\text{PartUnit } T := \nabla : R \rightarrow T \rightarrow R$, $R \sqsubseteq T$ (539); it is a unit when moreover every object carries an entire arrow to it, $\text{IsUnit } T := \text{PartUnit } T$ (545). A unitary Allegory is an allegory with one (548).	already there, for free . The monoidal unit I is the unit object, $I \circ a = 1$ is the entire arrow every object carries to it – (40) says so – and $\tau = !$; $?$ is maximal by Freyd's <code>Diag.«s_top»</code> . So every cartesian bicategory of relations is unitary, with nothing assumed.
pre-tabular $\$2.165$	f, g tabulate R when both are maps, $R = f^* g^*$, and $f^* f^* \cap g^* g^* = 1$ (Tabulates, 5502); R is tabular when some pair tabulates it (5506). An allegory is <code>PreTabularAllegory</code> when every morphism is contained in a tabular one (556). $\$2.165$: “If A is unitary then pre-tabularity is equivalent to the existence of tabulators for maximal morphisms”.	this is what $\otimes$ is . Given unitarity, the condition reduces to tabulating each Δ, ∇ , and a tabulation of τ, ι is precisely an object $a \otimes b$ with its two projections – the monoidal product Def. 4.1 assumes. That is the whole content of the swap: Freyd derives the product from tabulations, Def. 4.1 posits it. Not built. Redubbing from tabulations and products in <code>Map(A)</code> is <code>CartBicatOfUnitaryPretabular</code> ; <code>diag/PLAN</code> , <code>md</code> holds it back for want of a consumer.

Together they are the whole gap: **cartesian bicategory of relations** $\equiv$ **unitary pre-tabular allegory** (`DiagrammaticAlgebraofFirstOrderLogic` §10, after CartBicat–Walters). One half of that equivalence is `allegoryofCartBicat` and is a theorem in this repo; the other half is the two rows above.

6. Unitary and residual – what phases 8 and 9 add

Δ and ∇ are past where Freyd's §2.11 list stops, and past what Def. 4.1 can build: they need a **biproduct** on top of the Frobenius structure (`diag/Tape.lean`, after `TapeDiagrams.pdf`) and a **second composition** with linear adjoints (`diag/F0.lean`, after `DiagrammaticAlgebraofFirstOrderLogic.pdf`). Both are built, so both draw.

The two notations say which layer a picture is in. An union is a **tape**: the rounded wrapper is the second monoidal product, and its $\circ$ open and close a branch a particle takes exactly one of – the same fork–run–join shape as a meet, on a different product. A residual is **dashed** box, because it is not a generator of the generators at all: $R^* \circ S^* \circ R$ lives in the second composition, and a solid box would claim it was drawable in the first.

statement	picture, from the Lean statement
$R \sqsubseteq R \cup S = \2.2 in Freyd; here <code>Freyd.Diag.F0Cub.residual_left</code> , union $R S := (R, S) ; [1, !]$ (<code>diag/Tape.lean:84</code>); offer both branches, then forget which was taken.	
$R \cup S = S \cup R =$ Freyd, <code>Diag.union_comm</code> . The two branches of a tape are interchangeable, exactly as the two strands of Δ are.	
$I \cup R = R =$ Freyd, <code>Diag.Biproduct_bot_union</code> . but $a \otimes b := \text{toZero } a \circ (\text{toZero } b)^*$ (<code>diag/Tape.lean:87</code>), routed through the zero object; the exporter has no shape for it, so it prints as a box.	
$R (S \cup T) = R S \cup R T =$ Freyd, <code>Diag.union_composition</code> , distributes over union, which it does not. Freyd §2.2 distributivity, the reason a distributive allegory is more than an allegory.	
$(R \cup S)^* = R^* \cup S^* =$ Freyd, <code>Diag.residual_union</code> . The converse is a Δ -functor for the tape layer too.	
$R \cap (S \cup T) = (R \cap S) \cup (R \cap T) =$ Freyd, <code>Diag.F0Cub.residual_right</code> , the law that makes <code>distributiveAllegoryOfF0Cub</code> a distributive allegory (§2.2). The one picture with both layers in it: it meets inside a tape on the left, tapes around meets on the right.	
$(R / S) \sqsubseteq R =$ Freyd, <code>Diag.ClosedLinearBicat.«residual_comp_s»</code> ; the residual is a closed bound of the arrows it is the greatest of. Freyd's §2.2 division R / S is the largest T with $T S \sqsubseteq R$, and le_residual_l_iff (<code>diag/F0.lean:170</code>) is that Galois balancing in the full composition.	

7. Lemma 4.2 (ii) – the proof, not just the statement

`diag-export --proof draws a proof`. A calc chain elaborates to nested τ 's, and its intermediate terms are all recoverable from the proof term: flatten the spine, infer the type of each lead, draw one canvas per step. Nothing is replayed and nothing is retold – nine rows below are the chain the Lean kernel accepted for Freyd, `Diag.CartBicat.conv_comp`, which is Lemma 4.2 (ii), ($R : S$) * $= R^* S^*$.

7.1. First: why these pictures are not pictures of a converse

They are not, and that is the thing to get straight before reading them. The converse is drawn with two bends – a cup on the way in and a cap on the way out, each a pair of dots, $?$: Δ and $!$: ∇ – which is the left-hand figure below. The proof never draws that. `conv_unique` says an arrow is the converse of R as soon as it satisfies one equation between UNBENT arrows $a \otimes b = 1$: two strands in on the left, one cap on the right. Bending is a bijection (`bend_unbend/unbend_bend`), so settling the unbent equation settles the bent one, and the cap never has to be drawn at all.

That is why every row below has a cap and no cup, and why an arrow that “is a converse” shows up not as a bent box but as a box sitting on the bottom strand, mirrored.

the converse, as the paper draws it	the one rule the proof uses, twice
R^* is R with both wires bent round: in at the bottom left, cup at the top left, cap at the bottom right. Two dots at each bend, as <code>functorialSemanticsForRelationalTheories.pdf</code> draws it – a cup is $?$ then Δ , a cap is $!$. Transcribed by hand; a definition has no statement to export.	<code>conv_slide</code> (<code>diag/CB.lean:449</code>): a box on the BOTTOM strand facing a cap is the same as that box on the TOP strand, upright. Sliding round the bend is what turns R^* into R . This is the entire content of Lemma 4.2 (ii); the other seven steps are bookkeeping.

7.2. The chain

One row per theorem, not per step: the right-hand side of every step is the left-hand side of the next, so drawing steps would draw every picture twice. The first column is the term the Lean has it, fully bracketed, with $\circ$ binding tighter than $\cap$ – it comes off the same walk as the picture, so the two cannot drift apart. What which **strand** each box is on: bottom means mirrored, top means upright.

term	picture	the rule that reaches it
$\text{cap} : (S^* ; R^*) ; 1$		the start . What <code>UNIQUE</code> asks about S^* : both boxes on the bottom strand, mirrored – that is what “these are converses” looks like once the wires are unbent.
$= 1 \otimes S^* ; (1 \otimes R^* ; \text{cap})$		1. $\otimes$ is a functor , so doing two things to one strand in sequence is the sequence of doing them. Same picture.
$= 1 \otimes S^* ; (R \otimes 1 ; \text{cap})$		2. <code>conv_slide</code> . R^* is the box nearest the cap, so it slides round the bend: it leaves the bottom strand, arrives on the top, and comes back upright as R . The first of the two steps that prove anything.
$= (1 \otimes S^* ; R \otimes 1) ; \text{cap}$		3. <code>simp</code> only [<code>cat.assoc</code>], Re-bracketing – visible in the term column, invisible in the picture. Same picture.
$= R \otimes S^* ; \text{cap}$		4. <code>tension_split</code> – interchange. R is on the top strand and S^* on the bottom, so running them side by side are the same. Shows as a re-spacing only because the exporter lays a diagram out left to right.
$= (R \otimes 1 ; 1 \otimes S^* ; \text{cap})$		5. <code>tension_slide</code> – the same law, but the other way. Split them apart again. R first, so that S^* is now the box facing the cap, which is what the next rule step needs.
$= R \otimes 1 ; (1 \otimes S^* ; \text{cap})$		6. <code>simp</code> only [<code>cat.assoc</code>], Re-bracketing. Same picture.
$= R \otimes 1 ; (S \otimes 1 ; \text{cap})$		7. <code>conv_slide</code> , on S this time. S^* slides round the bend and arrives upright on the top strand. Both wires boxes are now on top both upright, in the order then S . The second and last rule step.
$= (R \otimes 1 ; S \otimes 1) ; \text{cap}$		8. <code>simp</code> only [<code>cat.assoc</code>], Re-bracketing. Same picture.
$= (R ; S) \otimes 1 ; \text{cap}$		9. <code>tension_comp</code> – interchange once more: the two top-strand boxes merge into one. This is $R ; S$; unbent, so <code>conv_unique</code> reads off $S^* R^* = (R ; S)^*$. Done.

What the **chain measures**, reads down the pictures and most consecutive pairs are the same drawing – every step, 1, 3, 6, 8, and 9 change nothing at all, and chain 4 changes only shape – while the term column beside them changes as they row. Those are associativity, the unit laws and interchange – precisely the language a string diagram has built into its geometry rather than as rewrites. Seven of these nine steps are bureaucracy the paper's draws do not charge for, and their theorem are two applications of `conv_slide`.

How this differs from the paper's proof. The paper proves (ii) by inserting a snake in the middle of R ; S and reading the two halves off as S^* and R^* – three pictures, all bent. The Lean never bends. Same theorem, different route. And the export shows the route the kernel accepted. Drawn by hand it would have shown the paper's three pictures and quietly implied those were what the repo had checked.

8. The modular law is proved, drawn

Freyd adjoints the modular identity: an allegory has no structure to derive it from, so it goes in as an axiom. Here it is a theorem, and this is its proof – Freyd, `Diag.modular_of_frobenius` (`diag/CB.Allegory.lean:48`), four terms, of which exactly one step is an inequality.

the modular law, $R S \cap T \leq (R \cap T S^*) S$	picture	the rule that reaches it
$(R ; S) \cap T$		the start : $R^* R \sqsubseteq R$. A meet, so a copy and a merge with R above and T below.
$= (\Delta ; R \otimes T) ; (S \otimes 1 ; \nabla)$		Factor out the shared prefix . ($R S$) $\circ$ T is ($R \otimes T$) ; ($S \otimes 1$), so everything before the S is common to both sides and can be set aside. An equality, and drawn as one.
$\leq (\Delta ; R \otimes T) ; (1 \otimes S^* ; (\nabla ; S))$		«∇ slide_conv , under the shared prefix, S slides through the merge and reappears as S^* on the other strand. The only inequality in the proof – everything else is reshaping.
$= R \cap T ; (S^* ; S)$		Hold the prefix back in $R \otimes (T S^*)$ is ($R \otimes T$) ; ($1 \otimes S^*$) , and the copy-meet pair reads as a meet again. ($R \cap T S^*$) $\sqsubseteq S$, which is the goal.

8.1. Inside the middle step

`«slide_conv»` (`diag/CB.lean:560`) is where the mathematics is, and it mentions only S – R and T sit untouched in the prefix throughout. Read on its own it says one arrow slides through a merge:

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S occurs once on the left and twice on the right. That is the tell: the lax copy inequation (42) is the only law in Def. 4.1 that may duplicate a box, so it has to be the inequality step – and its own chain shows it is the only one.

« ∇ slide_conv», $(S \otimes 1) ; \nabla \leq (1 \otimes S^*) ; \nabla ; S$	picture	the rule that reaches it
$S \otimes 1 ; \nabla$		the start : ($S \otimes 1$) ; ∇ .
$= (S ; \Delta) \otimes 1 ; 1 \otimes \text{cap}$		«∇ of cap» , backwards. The merge is rebuilt from a copy and a cap, which puts the term into the shape ($S ; \Delta$) $\otimes 1$ that (42) applies to.
$\leq (\Delta ; S \otimes S) \otimes 1 ; 1 \otimes \text{cap}$		(42), <code>lax_Δ</code> , $S ; \Delta$ becomes Δ : ($S \otimes S$) $\leq$ the box is copied. The whole of the modular law, in one step.
$= 1 \otimes S^* ; (\nabla ; S)$		«cap_tens_∇» and <code>conv_slide</code> . Reshaping back, and the surviving duplicate slides round the cap to become S^* – the same rule Lemma 4.2 (ii) is made of.

What the **associators cost**, $\otimes$ and p have no shape in this calculus and print as opaque boxes, so the two middle terms of the inner chain are less legible than its ends. That is the honest price of drawing what the Lean says: the reshaping steps really are that fiddly, and a hand-drawn picture would have hidden them by working up to coherence.

9. Maps: SV, TOT, INJ,